Name / Имя: _____ **Class:** _____ **Date:** _____

Laboratory Work No. 2 · Drops on a coin

Aim: observe how water molecules stick together (cohesion) and stick to the coin (adhesion).

Materials	Safety
• a glass of water • a dropper (pipette) • a coin • a paper towel	• Keep water away from sockets and devices. • Wipe spills at once. • Wash your hands after the lab.

1 · Pre-lab questions · Вопросы до работы

a) What is cohesion?

b) What other properties does water have?

2 · Predict · Предскажи

I predict _____ drops will stay on the coin before the water spills, because _____

3 · Procedure · Ход работы

1. Put the coin on a flat surface. Dry it with the towel.
2. Fill the dropper with water.
3. Slowly drip water onto the coin — **one drop at a time**.
4. Count the drops. Write the number when the coin is **full**.
5. Keep dripping. Write the number when the water **spills**.
6. Dry the coin. Repeat (trial 2, trial 3). Swap roles.

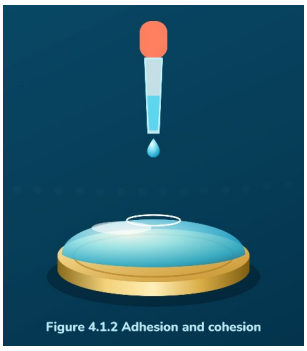


Figure 4.1.2 Adhesion and cohesion

4 · Results · Результаты

Write the **real** number of drops you counted.

Trial	Drops when the coin is full	Drops when the water spills
1		
2		
3		
Mean (average)		

Class results (from the board): lowest _____ highest _____ mean _____

Draw the water on the coin from the side. Label: coin, water, surface.

5 · Post-lab questions · Вопросы после работы

1. Why did the water on the coin not spill?

The water did not spill because _____

2. In which processes is this property used? How does it help living organisms?

3. What would happen if water molecules could not stick to each other?

If water molecules could not stick together, then _____

4. Why were the numbers of different pairs not the same? Give one reason.